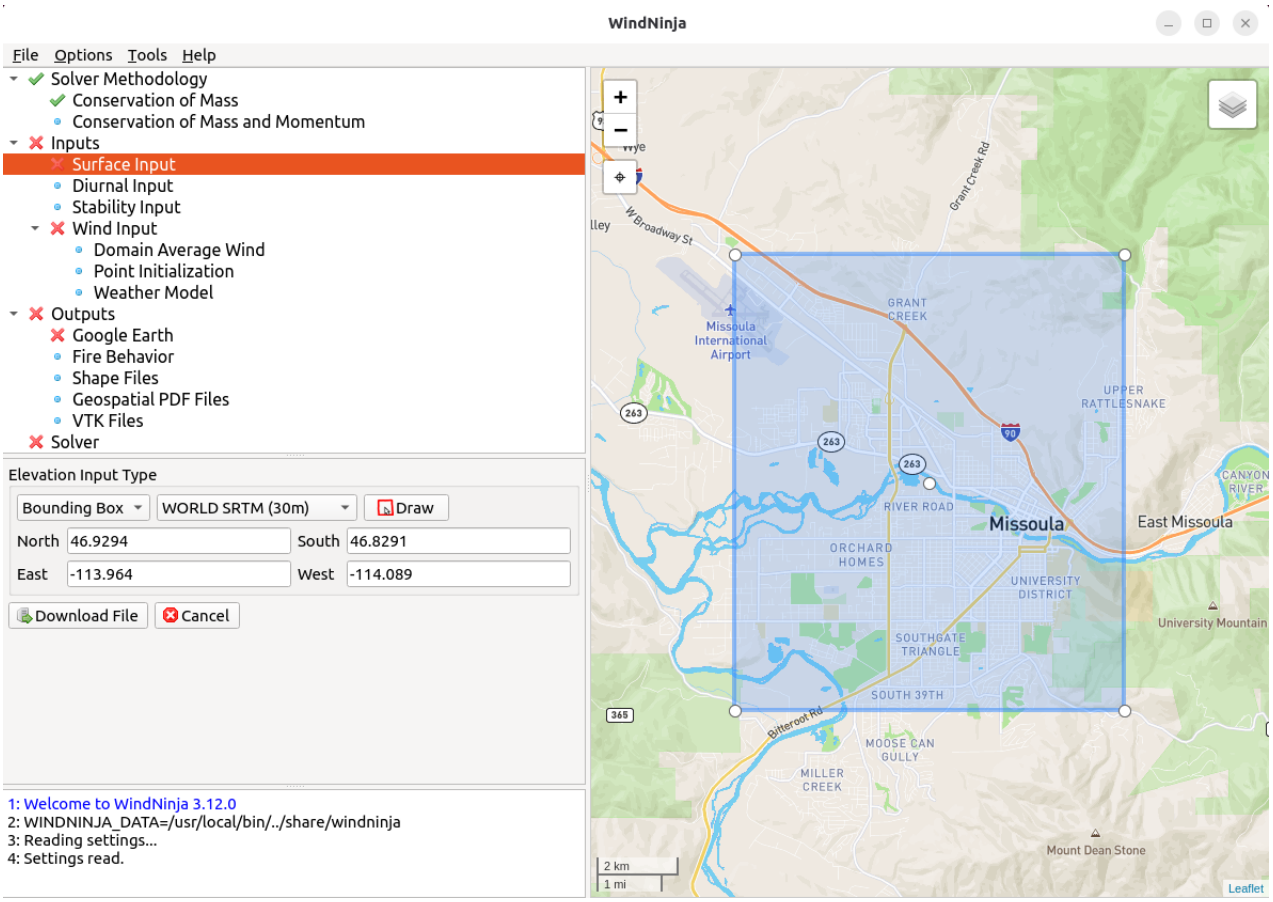
How to download an elevation file using WindNinja

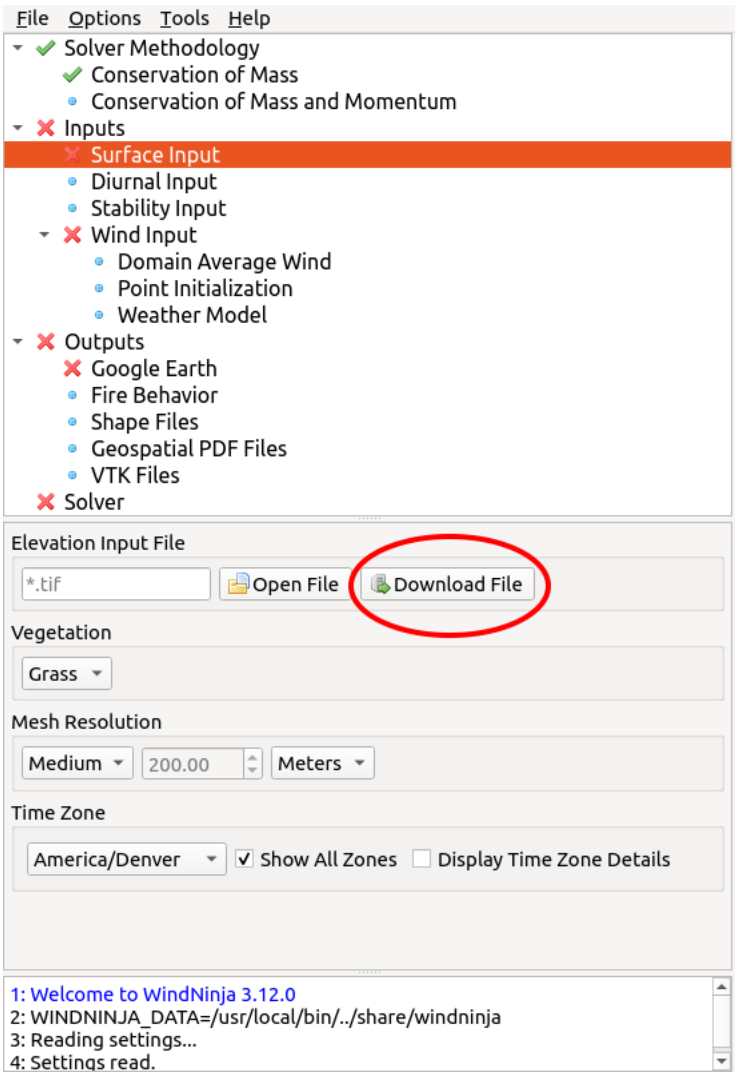


6/30/2023

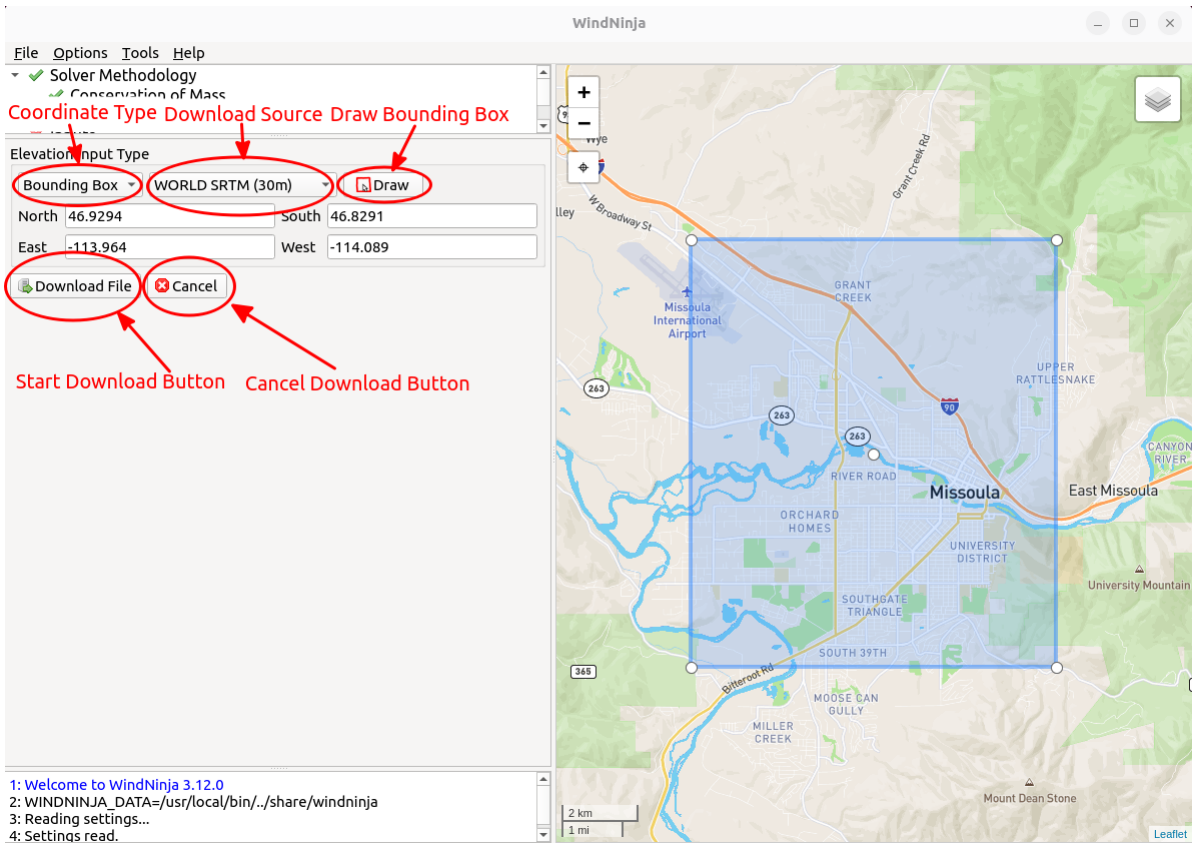
Introduction

This document describes the functionality built into WindNinja to download elevation files for wind modeling. We call this utility the "elevation file grabber". It allows you to use a custom ©Mapbox ©OpenStreetMap interface to zoom into and select a desired area for wind modeling. The elevation data for this area is downloaded by WindNinja from a USGS server and saved to a file on your computer. There is also a command line version of the elevation file grabber that is described here. The specific options available and work flow are described below.

To open the elevation file download page, start WindNinja and click *Surface Input* in the navigation tree at the top of WindNinja, then click the *Download File* button as shown below.



This area to the right of the navigation tree is an interactive mapping display used to navigate into the desired wind simulation area. There are a couple of ways to pan and zoom the map. Using a mouse, you can pan with the left button and zoom using a mouse scroll wheel. The keyboard can also be used to pan (arrow keys) and zoom (plus and minus keys).



Defining Your Simulation Area

There are three ways to define the area you want to download.

1. Click the *Draw* button in the upper right corner of the download page. Draw a box on the map bounding the area that you want to simulate. A transparent blue box will be drawn in the graphics window showing the area you've defined. The white points can be used to adjust the shape and location of the drawn bounding box.

2. Specify the *Bounding Box Coordinates* directly by entering the north and south latitudes and the east and west longitudes in decimal degrees that define your bounding box.

3. Use the *Point and Radius* method. Enter a latitude and longitude in decimal degrees specifying the center of the domain and a radius in miles specifying the distance from the center of the domain to the bounding box edge.

Once the bounding box has been defined, the coordinates will be displayed in the *Bounding Box* or *Point and Radius* text boxes depending on which is selected. These coordinates can

be edited to change the size of the bounding box, updating the blue box. If a bounding box has been already defined, clicking on the *Draw bounding box* button will delete it.

Elevation Data Source

There are 3 possible on-line data sets to download elevation data from. These are listed in the *Data Source* drop-down box at the top left of the graphics window. The main difference between these data sets is the spatial resolution and extent. The table below briefly describes these differences. You should be sure to choose a data set that covers your simulation area.

Data set name	Resolution	Spatial Extent	Description
WORLD SRTM	30 meters	Between approximately -60 and +60 latitude	Data from the Shuttle Radar Tomography Mission (SRTM)
WORLD GMTED	250 meters	Between approximately -60 and +85 latitude	Global Multi-resolution Terrain Elevation Data 2010 (GMTED 2010)
Landscape (LCP)	30 meters	Contiguous US, Hawaii, PR, Southern Alaska	FARSITE Landscape files including vegetation information

Download The File

Once you have defined your download area and selected the elevation data source, click the *Download File* button at the bottom left of the download page to download the elevation file. A window will open to allow you to specify a file name and location. The downloaded file will be a GeoTIFF file (*.tif) in best fit UTM projection using WGS84. After the file is successfully downloaded, the downloaded file will be automatically loaded into the Surface Input page.